



Siddaganga Institute of Technology, Tumkur

Department of Electrical and Electronics Engineering

Interdisciplinary Project Phase-I Presentation

on

"Green Synthesis of TiO_2 nanoparticles and development of PLA- TiO_2 nanocomposite for electrical insulator applications"

by

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- Motivation
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Environmental pollution caused by non-biodegradable petroleum-based plastics has increased the demand for eco-friendly and biodegradable materials. Polylactic Acid is a biodegradable and bio-based polymer widely used in packaging, biomedical, and engineering applications. However, pure PLA has some limitations such as:

- 1.Low thermal stability
- 2.Brittleness
- 3.Poor UV resistance
- 4.Limited mechanical strength

Electrical insulation is important in electronic and power systems. Normal plastic materials are harmful to the environment because they are non-biodegradable. PLA is an eco-friendly biodegradable plastic, but it has low heat and electrical strength. To improve its properties, Titanium Dioxide (TiO_2) nanoparticles are added.

In this project, TiO_2 nanoparticles are made using a fast and energy-efficient method called solution combustion synthesis (SCS).

- To synthesize TiO_2 nanoparticles using a controlled Solution Combustion Process.
- To optimize the calcination temperature($600^{\circ}C-650^{\circ}C$)for phase purity(Anatase/Rutile)
- To develop a PLA- TiO_2 nanocomposite suitable for electrical insulation or capacitive applications.
- To evaluate the morphological and structural characteristics of the synthesized nanoparticles.

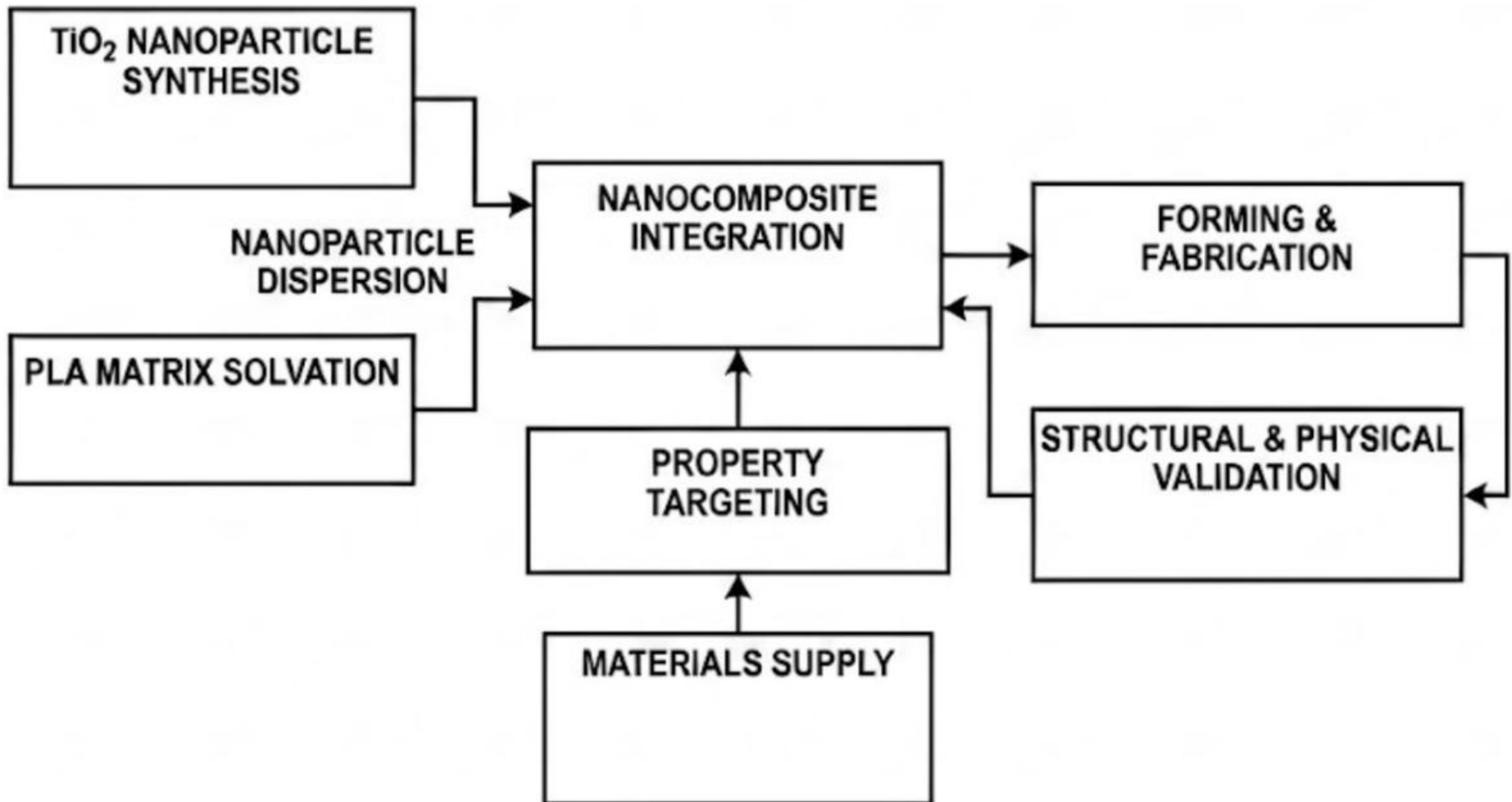
Author(s), Conference/Journal, Year	Title of Paper	Remarks
S. T. Aruna and A. S. Mukasyan Current Opinion in Solid State and Materials Science, 2008	“Combustion synthesis and nanomaterials”	Highly effective, rapid, and low-energy chemical pathway for crystalline nanopowders, but causes agglomeration due to high local reaction temperatures.
K. Nagaveni, G. Sivalingham, M. S. Hegde, and G. Madras Environmental Science & Technology, 2004	“Photocatalytic Degradation of Organic Compounds over Combustion-Synthesized Nano-TiO ₂ ”	Successfully achieves high surface area and excellent crystallinity in nano-TiO ₂ , but focuses only on chemical reactivity , leaving electrical/dielectric capabilities.
V. Senthilkumar and P. Vickraman Applied Surface Science, 2011	“Effect of calcination temperature on the structural and optical properties of TiO ₂ nanoparticles”	Provides precise phase-evolution tracking from anatase to rutile, but requires energy-intensive, long-duration furnace calcination , which elevates manufacturing costs.

Author(s), Journal, Year	Title of Paper	Remarks
H. M. Ahmed et al. Materials 2023	A Novel Polyester Varnish Nanocomposites for Electrical Machines with Improved Thermal and Dielectric Properties Using Functionalized TiO ₂ Nanoparticles	Improves thermal and dielectric performance using TiO ₂ nanoparticles, but focuses on polyester varnish rather than PLA matrices.
M. Zorah et al. JARFMTS 2024	Improvement of Thermomechanical Properties of Polylactic Acid via Titania Nanofillers Reinforcement	Enhances mechanical strength and thermal properties of PLA, but does not investigate electrical insulation properties.
F. Riyanti et al. Turkish Journal of Chemistry 2022	The Effect of Fuel on the Physicochemical Properties of ZnFe ₂ O ₄ Synthesized by Solution Combustion Method	Shows the importance of fuel selection in combustion synthesis, but focuses on ZnFe ₂ O ₄ rather than TiO ₂ composites.

- Solution Combustion Synthesis (SCS) is established as an energy-efficient, rapid pathway for producing highly crystalline ceramic oxides .
- A major limitation* identified in standard TiO_2 synthesis is the reliance on high-temperature, long-duration furnace calcination, which elevates manufacturing costs.
- A prominent application gap exists: While TiO_2 synthesis is well-documented, its behavior as a reinforcing dielectric filler within a fully biodegradable biopolymer matrix is virtually unexplored.
- Proposed solution: A sustainable, high-performance TiO_2 nanocomposite
- designed specifically for electrations.
- This approach addresses identified gaps by utilizing a rapid, low-energy urea combustion route to generate high-purity TiO_2 and integrate it into a Polylactic acid (PLA) host, effectively upgrading an eco-friendly biopolymer into a robust engineering dielectric.

This makes the TiO_2 nanocomposite both sustainable and technically optimized compared to existing work.

Block diagram description



The PLA-TiO₂ nanocomposite is made in three steps. First, TiO₂ nanoparticles are prepared using a chemical process. Then, they are mixed with PLA biopolymer to improve its properties.

Finally, the material is tested for strength, heat resistance, and electrical insulation. This helps create an eco-friendly insulating material.

Hardware system:

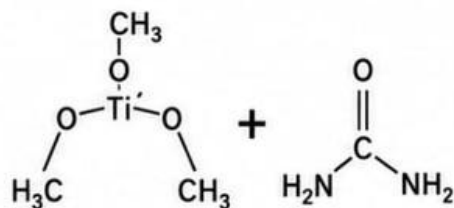
For analysis: To verify that the synthesis was successful, an X-Ray Diffraction (XRD) system is employed.

Software system:

The XRD software controls the hardware, collects diffraction data, and processes it using mathematical algorithms. It analyzes the diffraction patterns to determine the crystal structure and properties of the synthesized TiO₂ nanoparticles.

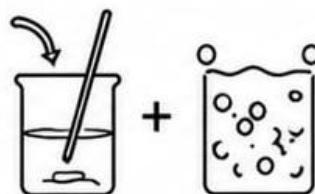
SECTION 1: TiO₂ NANOPARTICLE SYNTHESIS

1.1 PRECURSORS (Solution Combustion Method)



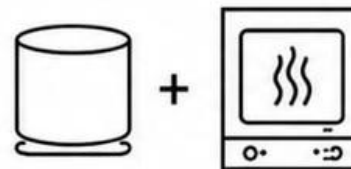
- Ti{OCH(CH₃)₂}₄
- Urea (fuel)
- Dissolved in DI Water

1.2 HYDROLYSIS & CONDENSATION



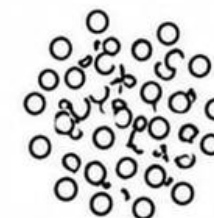
Mixing,
Stirring,
Hydrolysis & Condensation
Gel formation

1.3 DRYING & CALCINATION



Drying gel,
Heating in furnace
450–600 °C
(Calcination)

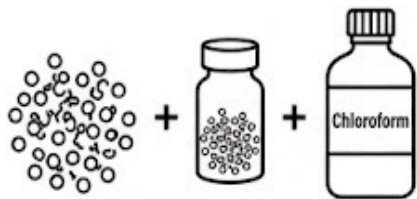
1.4 TiO₂ NANOPARTICLES



TiO₂ NANOPARTICLES

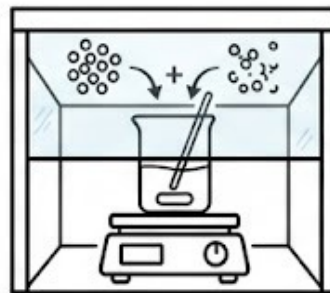
SECTION 2: PLA-TiO₂ NANOCOMPOSITE FABRICATION (SOLUTION CASTING)

2.1 PRECURSORS & SOLVENT (Solution Casting Components)



- Polylactic Acid (PLA) matrix
- Synthesized TiO₂ Nanoparticles (1%, 3% wt.)
- Chloroform (acting solvent)

2.2 DISSOLUTION & DISPERSION



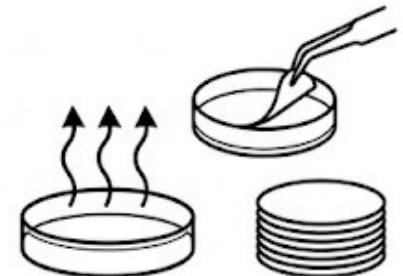
- Dissolving PLA in Chloroform
- Adding TiO₂ nanoparticles
- Continuous magnetic stirring for homogeneous dispersion

2.3 SOLUTION CASTING



- Transferring liquid matrix to casting substrates
- Ensuring uniform thickness distribution

2.4 SOLVENT EVAPORATION & FILM FORMATION



- Controlled evaporation of Chloroform at room temperature
- Peeling of solid nanocomposite films

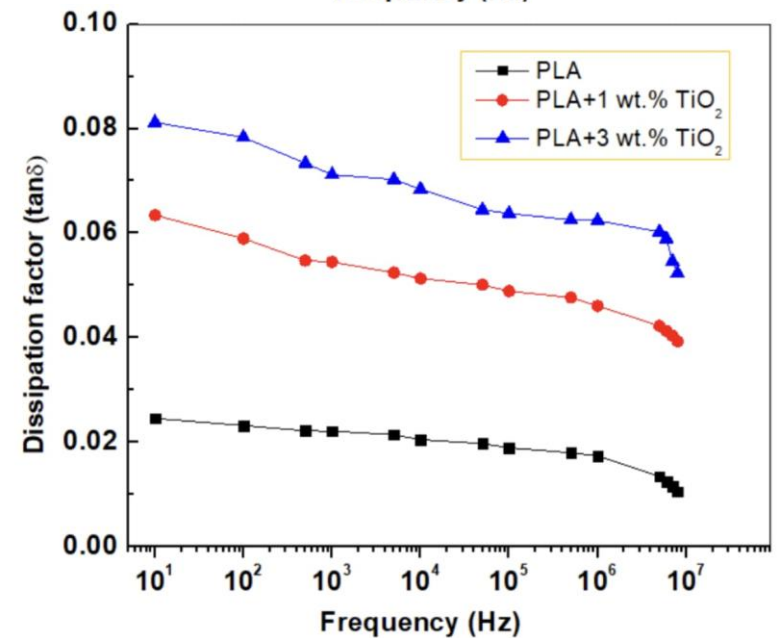
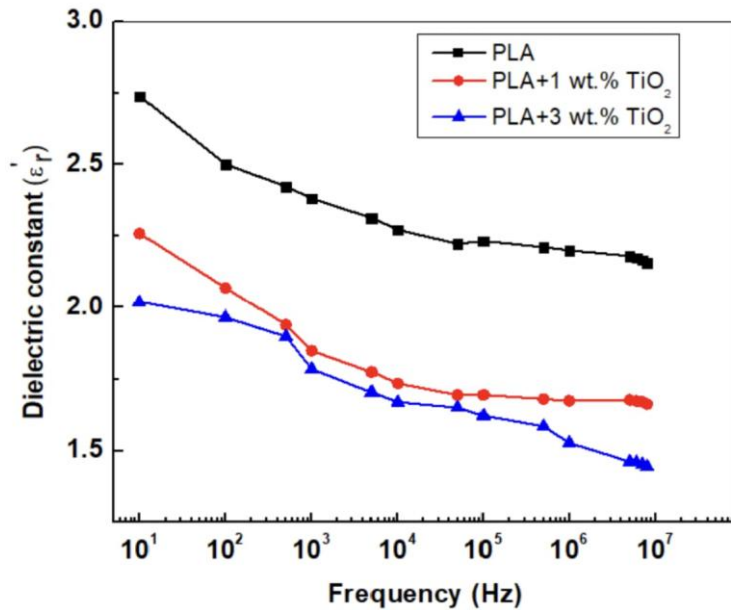
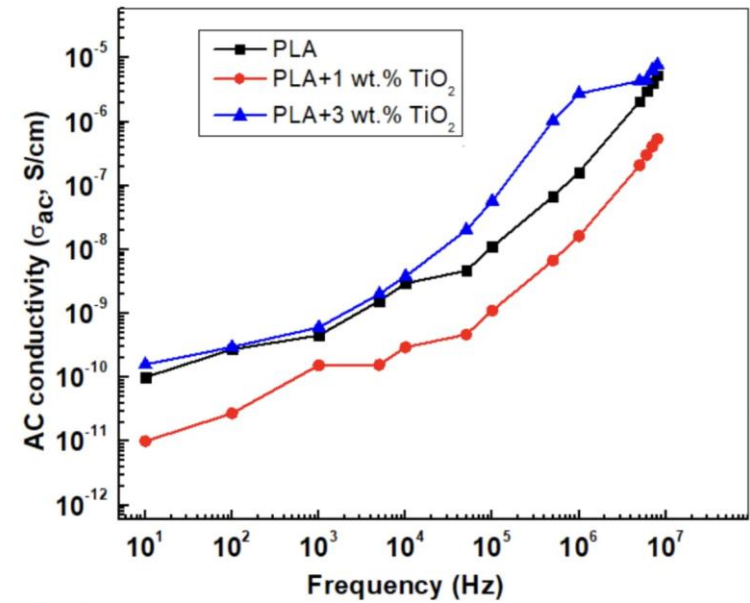
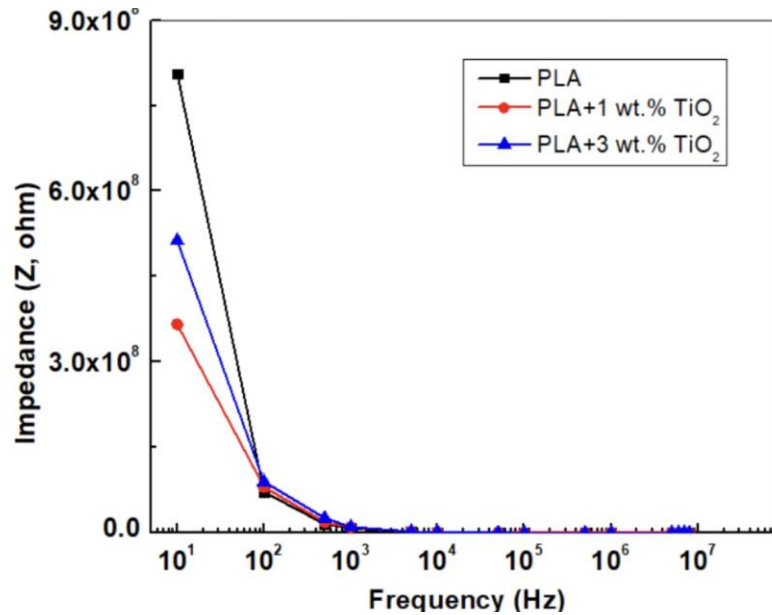
Solid PLA-TiO₂ Nanocomposite Films ready for testing.

The study confirms that incorporating ceramic nanofillers into polymer matrices significantly enhances composite performance. Among different compositions, optimized filler loading produced the best balance of:

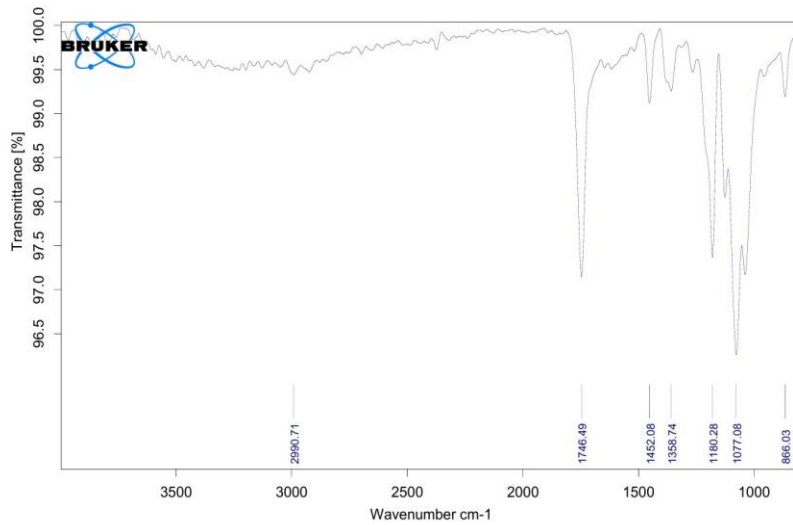
1. Thermal stability
2. Mechanical strength
3. Dielectric performance
4. Environmental sustainability

Therefore, polymer–filler composites, especially PLA–TiO₂ nanocomposites, show strong potential for next-generation eco-friendly electrical insulation applications.

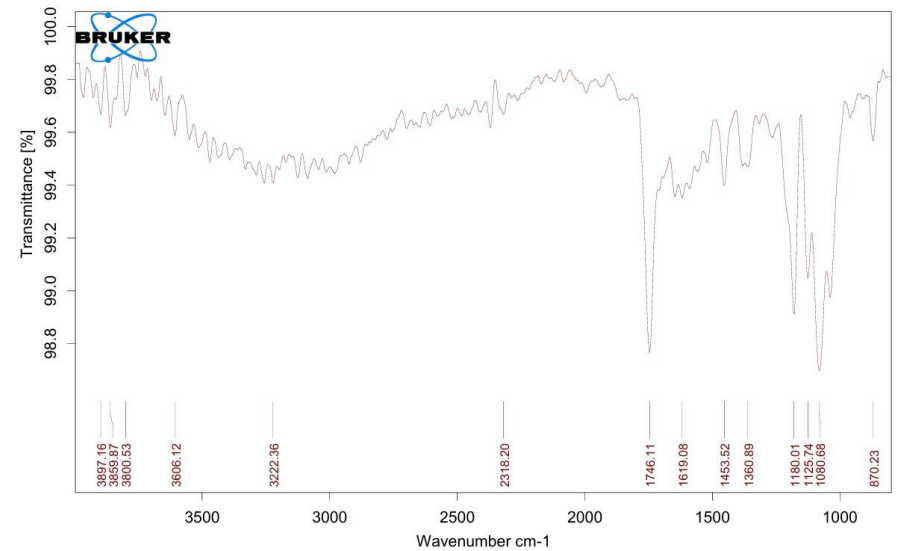
Results



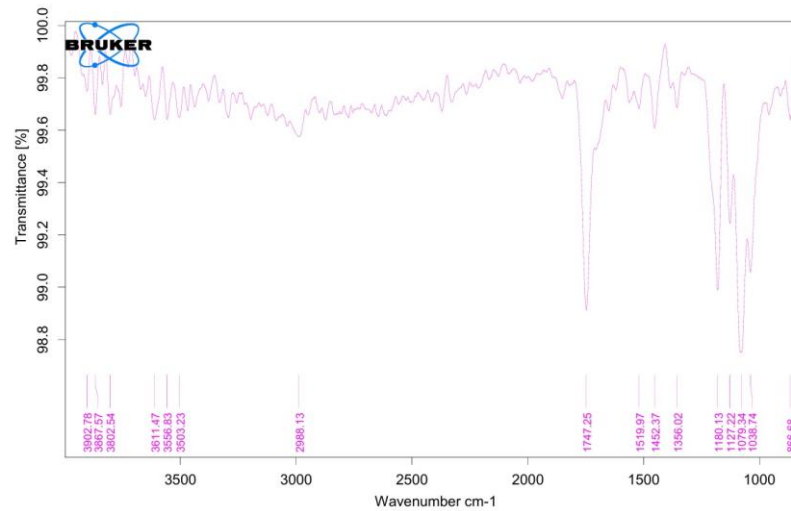
Results



FTIR graph of 1% TiO₂

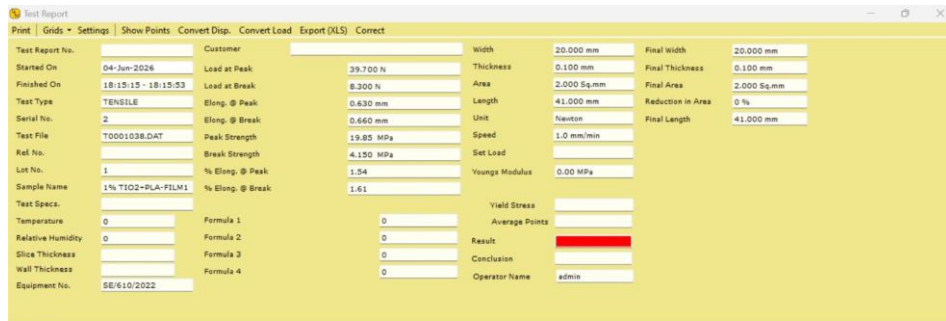


FTIR graph of 3% TiO₂

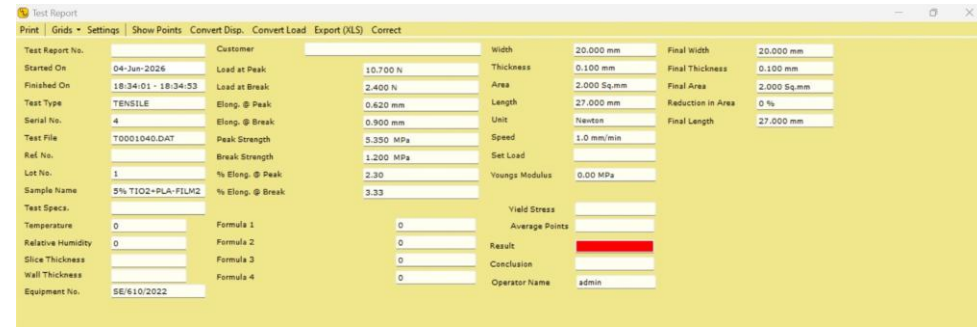


FTIR graph of Pure PLA

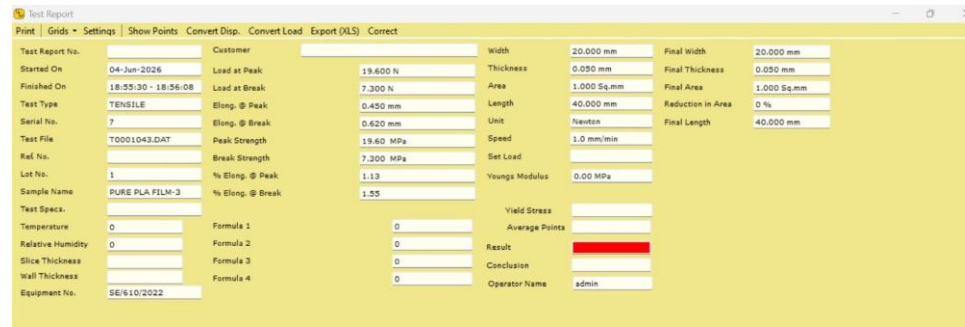
Results



Mechanical Tensile report of 1% TiO₂



Mechanical Tensile report of 3% TiO₂



Mechanical Tensile report of Pure PLA

1. Literature Review Completed
2. Analyzed limitations of neat PLA
3. Investigated the role of ceramic nanofillers such as TiO_2
4. Study of TiO_2 Nanoparticle and done with synthesis
5. Made the PLA- TiO_2 nanocomposite
6. Material characterization and testing techniques
 - XRD:** Analyzes crystal structure and particle size.
 - FTIR:** Identifies chemical bonds and functional groups .
 - Mechanical Test:** Measures strength and mechanical performance.
 - Electrical (LCR) Test:** Evaluates high-frequency insulation stability and dielectric loss.

Plan of action



Events	March	April	May	June
Problem Formulation and Synopsis Preparation	✓			
Literature survey for possible methods to address the problem	✓			
Design solution		✓		
Identification of platforms/tools/devices/controllers		✓		
Implementation		✓		
Validate the design, analyze and interpret the results			✓	
Preparation of the report			✓	
Demonstration				✓

Budget analysis



In this section the total estimated budget for completing the project has to be mentioned

Components	Cost in Rs.
1. Titanium Isopropoxide (TTIP)	1080/-
Total	1080/-

- [1] S. T. Aruna and A. S. Mukasyan, “**Combustion synthesis and nanomaterials,**” Current Opinion in Solid State and Materials Science, vol. 12, no. 3-4, pp. 44-50, Dec. 2008.
- [2] K. Nagaveni, G. Sivalingham, M. S. Hegde, and G. Madras, “**Photocatalytic Degradation of Organic Compounds over Combustion-Synthesized Nano- TiO_2 ,**” Environmental Science & Technology, vol. 38, no. 5, pp. 1600-1604, Mar. 2004.
- [3] V. Senthilkumar and P. Vickraman, “**Effect of calcination temperature on the structural and optical properties of TiO_2 nanoparticles,**” Applied Surface Science, vol. 257, no. 7, pp. 2970-2975, Jan. 2011.

- [4] F. Riyanti, W. Purwaningrum, N. Yuliasari, S. Putri, N. Aprianti, and P. L. Har iani, **“The effect of fuel on the physiochemical properties of ZnFe_2O_4 synthesized by solution combustion method,”** Turkish Journal of Chemistry, vol. 46, no. 6, pp. 1875–1882, 2022
- [5] M. Zorah, I. R. Mustapa, N. Daud, J. H. Nahida, N. A. S. Sudin, A. A. Majhool, and E. Mahmoudi, **“Improvement of thermomechanical properties of polylactic acid via titania nanofillers reinforcement,”** Journal of Advanced Research in Fluid Mechanics and Thermal Sciences, vol. 70, no. 1, pp. 97–111, 2024
- [6] H. M. Ahmed, N. M. K. Abdel-Gawad, W. A. Afifi, D. A. Mansour, M. Lehtonen, and M. M. F. Darwish, **“A novel polyester varnish nanocomposites for electrical machines with improved thermal and dielectric properties using functionalized TiO_2 nanoparticles,”** Materials, vol. 16, no. 19, p. 6478, Sep. 2023.

Thank you